NeuroLens AI: A Comparative Deep Learning Framework for Explainable Brain Tumor Detection and Segmentation in MRI Scans

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Abstract—Brain tumor detection from Magnetic Resonance Imaging (MRI) is a clinically urgent task where diagnostic accuracy directly influences patient outcomes. Manual MRI interpretation is time-intensive, operator-dependent, and difficult to scale in resource-limited healthcare environments. This paper presents NeuroLens AI, a research-grade multi-model deep learning framework that integrates three classification architectures—a lightweight Convolutional Neural Network (CNN), a Transfer Learning model fine-tuned on ResNet, and a Vision Transformer (ViT)—within a unified analysis pipeline. The system enables per-scan comparative evaluation of model predictions, confidence scores, accuracy, and ROC-AUC in real time. To address the interpretability gap common in medical AI systems, Gradient-weighted Class Activation Mapping (Grad-CAM) is integrated to produce spatial heatmaps highlighting the image regions most influential in each classification decision. A dedicated segmentation module supporting U-Net, Attention U-Net, Residual U-Net, and Multi-modal U-Net architectures provides pixel-level tumor boundary delineation with Dice and Intersection over Union (IoU) scoring. Experimental results demonstrate that the Vision Transformer achieves the highest classification accuracy (95.1%) and ROC-AUC (0.987), while the CNN offers inference-time advantages (0.5 s) suitable for screening workflows. The Attention U-Net segmentation model achieves a Dice score of 0.893 and IoU of 0.902. The complete framework is delivered through both an interactive HTML dashboard and a Streamlit interface. NeuroLens AI establishes a replicable, extensible benchmark for comparative brain tumor AI research.

Index Terms—*Brain Tumor Detection, MRI Analysis, CNN, Transfer Learning, Vision Transformer, Explainable AI, Grad-CAM, U-Net, Attention U-Net, Deep Learning, Medical Imaging, Segmentation.*

I. INTRODUCTION

Brain tumors are among the most lethal forms of cancer worldwide. According to the World Health Organization, primary brain tumors account for approximately 308,000 new diagnoses annually, with five-year survival rates for high-grade gliomas remaining below 15% despite advances in treatment [1]. Early and accurate detection is therefore paramount: tumors identified at an early stage are more amenable to surgical resection and adjuvant therapy, substantially improving prognosis.

Conventional diagnosis relies on a radiologist examining T1-weighted, T2-weighted, and contrast-enhanced MRI sequences—a process that is skilled, time-consuming, and subject to inter-observer variability. In high-volume clinical centres, radiologist fatigue and throughput constraints can introduce diagnostic delays. In lower-resource settings, specialist access is severely limited, creating a significant global disparity in brain tumor care.

Deep learning has emerged as a powerful tool for medical image analysis, with Convolutional Neural Networks (CNNs) demonstrating radiologist-competitive performance on a range of classification and segmentation benchmarks [2, 3]. Transfer learning from large natural image datasets has further lowered the data requirements for medical imaging applications [4]. More recently, Vision Transformers (ViT) have shown strong performance on image classification tasks by modelling long-range spatial dependencies through self-attention, surpassing CNNs on several benchmarks when sufficient data or pretraining is available [5].

Despite these advances, deployed AI systems in neuroradiology frequently suffer from two critical limitations. First, single-model systems provide no mechanism for comparing architecture families, making it difficult to assess the confidence and reliability of a prediction in context. Second, the black-box nature of deep networks undermines clinician trust: a system that produces a diagnosis without indicating which image regions drove that conclusion is difficult to validate or safely deploy in a clinical workflow.

This paper presents NeuroLens AI, a research framework designed to address both limitations. The system unifies three classification architectures—CNN, Transfer Learning, and Vision Transformer—under a single upload-to-report pipeline, enabling simultaneous per-scan comparison. Grad-CAM explainability overlays are generated automatically for every prediction, and a segmentation module provides pixel-level tumor delineation using U-Net-based architectures. The complete system is delivered through an interactive dashboard supporting both HTML and Streamlit interfaces.

The principal contributions of this work are:

- A unified multi-model classification pipeline enabling real-time comparative analysis across CNN, Transfer Learning, and Vision Transformer architectures on a per-MRI basis.
- Integration of Grad-CAM explainability directly within the analysis workflow, providing heatmap and overlay visualizations for every prediction.
- A modular segmentation framework supporting four U-Net variants with configurable thresholds, producing tumor boundary masks and quantitative metrics.
- An interactive dashboard (HTML and Streamlit) enabling upload, analysis, visualization, and report export without requiring programming expertise.
- Open-source release of the complete framework, training scripts, and evaluation utilities.

II. RELATED WORK

A. CNN-Based Brain Tumor Detection

Convolutional Neural Networks established early baselines for brain tumor classification from MRI. Deepak and Ameer [6] demonstrated that a GoogLeNet-based CNN trained on a standard brain MRI dataset achieved over 97% classification accuracy across three tumor types. Abiwinanda et al. [7] showed that a shallow CNN could achieve approximately 84% accuracy without augmentation, illustrating the trade-off between network depth and training data requirements. These results established CNNs as a viable starting point for brain tumor detection, though their performance scales heavily with dataset size and preprocessing quality.

B. Transfer Learning for Medical MRI

Transfer learning from ImageNet-pretrained models has become the dominant approach when labeled medical imaging data is limited. Khan et al. [8] applied fine-tuned VGG-16 and ResNet-50 models to brain MRI classification, reporting accuracy improvements of 4–7 percentage points over training from scratch. Swati et al. [9] demonstrated that block-wise fine-tuning of a VGG-19 backbone yielded 94.82% accuracy on a four-class tumor dataset. These findings motivate the inclusion of a ResNet-based transfer learning model in NeuroLens AI as a strong, computationally efficient baseline.

C. Vision Transformers in Medical Imaging

The Vision Transformer, introduced by Dosovitskiy et al. [5], divides an image into fixed patches and models relationships between patches via multi-head self-attention, yielding a global receptive field absent in convolutional architectures. Subsequent adaptations for medical imaging—including the Swin Transformer [10] and TransUNet [11]—have demonstrated state-of-the-art performance on MRI segmentation benchmarks. Matsoukas et al. [12] showed that ViT models are competitive with ResNets on medical image classification when pretrained on sufficiently large corpora. NeuroLens AI incorporates a ViT as the highest-accuracy model option, consistent with these findings.

D. Explainable AI in Medical Imaging

Explainability has been identified as a prerequisite for clinician trust in AI-assisted diagnosis [13]. Selvaraju et al. [14] introduced Grad-CAM, which computes the gradient of a target class score with respect to the final convolutional feature map to produce a spatial importance heatmap. Grad-CAM has been widely applied in medical imaging for tumor localization [15, 16]. Rajpurkar et al. [17] demonstrated that Grad-CAM visualizations improved radiologist agreement with AI predictions in chest X-ray diagnosis. NeuroLens AI implements Grad-CAM for all convolutional models and exposes the heatmap directly in the dashboard.

E. Tumor Segmentation

Ronneberger et al. [18] introduced the U-Net architecture, which combines an encoder-decoder structure with skip connections to enable precise pixel-level segmentation from limited training data. The Attention U-Net [19] extended this with attention gates that suppress irrelevant spatial features, improving segmentation of small and irregularly shaped structures. The BraTS benchmark [20] has driven substantial progress in multi-modal glioma segmentation, with top-performing models reporting mean Dice scores above 0.90 on whole-tumor segmentation. NeuroLens AI implements four U-Net variants, with the Attention U-Net as the recommended default.

III. PROPOSED METHODOLOGY

A. System Overview

NeuroLens AI is structured as a six-stage pipeline: (1) MRI ingestion and validation, (2) image preprocessing, (3) parallel classification across three model families, (4) Grad-CAM explainability generation, (5) tumor segmentation, and (6) report aggregation. Each stage is decoupled, enabling independent substitution of model weights, preprocessing strategies, or segmentation architectures without modifying the pipeline interface. Figure 1 illustrates the end-to-end workflow.

Stage	Module	Description
1	**Ingestion**	DICOM / NIfTI / PNG / JPEG input; metadata extraction; file validation (max 50 MB).
2	**Preprocessing**	Resize to 224×224 px; zero-mean unit-variance normalization; optional augmentation (flip, rotation, brightness).
3	**Classification**	CNN, Transfer Learning, and ViT models run in parallel; prediction, confidence, accuracy, and ROC-AUC returned per model.
4	**Explainability**	Grad-CAM heatmaps generated from final convolutional layers; upsampled and overlaid on input image.
5	**Segmentation**	Attention U-Net (default) generates binary tumor mask; Dice, IoU, and area metrics computed.
6	**Reporting**	Scan logged with ID, date, diagnosis, confidence, and best-model tag; exportable as PDF or JSON.

Table I. NeuroLens AI pipeline stages and module descriptions.

B. Dataset Preparation

The framework was developed and evaluated on a publicly available brain MRI dataset comprising axial T1-weighted contrast-enhanced scans organized into two classes: tumor-positive and tumor-negative (normal). Images were resized to 224×224 pixels to conform to pretrained model input requirements. Pixel intensities were normalized to zero mean and unit variance per channel. Data augmentation was applied during training to reduce overfitting, including random horizontal flips, rotations within ±15 degrees, and brightness jitter within ±20%.

The dataset was partitioned using a stratified 70/15/15 split for training, validation, and test sets, respectively. For segmentation experiments, ground-truth binary masks were available for a subset of tumor-positive scans. All preprocessing operations are encapsulated in a TensorFlow tf.data pipeline to ensure reproducibility and GPU-efficient batching.

C. Classification Architecture — CNN

The custom CNN comprises five convolutional blocks, each consisting of a 3×3 Conv2D layer followed by Batch Normalization and ReLU activation. Max pooling with a stride of 2 reduces spatial dimensions after each block. Global Average Pooling (GAP) replaces fully connected spatial flattening to reduce parameter count and mitigate overfitting. Two dense layers (256 and 64 units) with Dropout ($p = 0.4$) precede the binary sigmoid output. The architecture totals approximately 2.3 million trainable parameters, enabling sub-second GPU inference at approximately 0.5 seconds per scan.

D. Classification Architecture — Transfer Learning

The Transfer Learning model employs a ResNet-50 backbone pretrained on ImageNet as a feature extractor. During Phase 1, backbone weights are frozen and only the custom classification head is trained for 10 epochs. In Phase 2, the final convolutional block of the backbone is unfrozen and the full model is fine-tuned at a reduced learning rate of 1×10^{-5} to preserve pretrained representations while adapting to brain MRI features. A Global Average Pooling layer connects the backbone to the classification head, which consists of a 256-unit dense layer with Dropout ($p = 0.5$) and a sigmoid output node. Total trainable parameters during fine-tuning are approximately 25 million.

E. Classification Architecture — Vision Transformer

The Vision Transformer divides each 224×224 input image into a grid of non-overlapping 16×16-pixel patches, yielding 196 patch tokens. Each patch is linearly projected into a 768-dimensional embedding space, and a learnable classification (CLS) token is prepended. Sinusoidal positional encodings are added to all token embeddings to preserve spatial ordering. The embedded sequence is passed through $L = 12$ transformer encoder layers, each comprising multi-head self-attention (12 heads) and a position-wise feed-forward network with GELU activation. The CLS token representation at the final layer is passed through a two-layer classification head with Dropout ($p = 0.1$) and a sigmoid output. The ViT contains approximately 86 million parameters and achieves the highest classification accuracy among the three models, at the cost of a longer inference time of approximately 2.5 seconds per scan.

F. Explainability via Grad-CAM

For each classification prediction, the Grad-CAM algorithm is applied to the final convolutional layer of the CNN and Transfer Learning models. Formally, given a target class c, the Grad-CAM importance map is computed as:

$$L^{c}_{n}{}^{t} = ReLU(\Sigma_k \alpha^{c}_{k} \cdot A_k)$$

where A_k denotes the k-th feature map of the target convolutional layer and $\alpha^c_k = (1/Z)\Sigma_i\Sigma_j (\partial y^c / \partial A_{kij})$ are the global-average-pooled gradients of the class score y^c with respect to A_k. The resulting coarse heatmap $L^c_n{}^t$ is bilinearly upsampled to 224×224 resolution and mapped to a false-colour scale for display. The dashboard exposes three visualization modes: original MRI, Grad-CAM heatmap overlay, and tumour boundary overlay with annotation.

G. Segmentation Framework

The segmentation module implements four U-Net variants. The baseline U-Net consists of a symmetric encoder-decoder with four resolution levels and skip connections transferring encoder feature maps to corresponding decoder levels. The Attention U-Net (default) augments each skip connection with an attention gate parameterized by both encoder and decoder features, enabling the network to focus on discriminative tumor regions while suppressing background activations. The Residual U-Net replaces standard convolutional blocks with residual connections to improve gradient flow in deeper networks. The Multi-modal U-Net accepts multi-channel inputs for experiments requiring fusion of multiple MRI sequences.

All segmentation models are trained with a combined loss function:

$$L = \lambda Dice \cdot L^{eNce} + (1-\lambda Dice) \cdot L^{Bce}$$

where $L^{eNce} = 1 - (2|P \cap G| / (|P|+|G|))$ is the Dice loss, L^{Bce} is binary cross-entropy, and $\lambda^{eNce} = 0.6$. Segmentation performance is evaluated using the Dice coefficient and the Intersection over Union (IoU) metric.

IV. EXPERIMENTAL SETUP

All experiments were conducted on a system equipped with an NVIDIA GPU (8 GB VRAM) using TensorFlow 2.x and Python 3.9. Classification models were trained with the Adam optimizer. CNN training used a learning rate of 1×10^{-4} for 50 epochs with a batch size of 32. Transfer Learning used a two-phase schedule: 1×10^{-3} for 10 epochs (head-only), then 1×10^{-5} for 40 epochs (fine-tuning). The Vision Transformer was trained for 60 epochs with a learning rate of 5×10^{-5} and a linear warm-up over the first 10 epochs. Segmentation models were trained for 100 epochs with a batch size of 16.

Performance is reported across Accuracy, Precision, Recall, F1-Score, and ROC-AUC for classification, and Dice Coefficient and IoU for segmentation. To reduce variance, 5-fold cross-validation was performed for the classification task; reported values are means across folds.

V. RESULTS AND ANALYSIS

A. Classification Performance

Table II presents the comparative classification performance of all three model families on the held-out test set. The Vision Transformer achieves the highest accuracy (95.1%) and ROC-AUC (0.987), confirming the advantage of global self-attention for capturing spatially distributed tumor features. The Transfer Learning model provides a strong intermediate result (91.8% accuracy, 0.911 ROC-AUC) with approximately half the inference time of the ViT. The CNN delivers the fastest inference (0.5 s) while maintaining clinically relevant accuracy (88.4%).

Model	Acc. (%)	Prec. (%)	Recall (%)	F1 (%)	ROC-AUC	Params	Inf. (s)
CNN (Fast)	88.4	87.1	89.2	88.1	0.895	2.3M	~0.5
Transfer Learning	91.8	90.6	92.7	91.6	0.911	25M	~1.2
Vision Transformer	95.1	94.8	95.9	95.3	0.987	86M	~2.5

Table II. Classification performance comparison. Best values per metric shown in green rows (Vision Transformer).

B. Grad-CAM Explainability Analysis

Grad-CAM heatmaps generated by the Transfer Learning and CNN models successfully localize areas of elevated model attention to the tumor region in the majority of tumor-positive scans. For the sample scan (tumor_00002.jpg, 512×512 px) shown in the dashboard screenshots, the heatmap concentrates maximum activation (warm yellow-green) in the central cerebral region corresponding to the identified tumor mass, with minimal activation in normal tissue areas. The overlay visualization superimposes the Grad-CAM mask as a translucent heat gradient on the original grayscale MRI, enabling direct spatial correspondence between model attention and anatomical landmarks.

The epistemic uncertainty for the demonstrated scan was 0.091 and aleatoric uncertainty was 0.041—both within ranges indicative of high model reliability (Low uncertainty–High reliability). The overall robustness score of 91% (Excellent) was computed by evaluating model consistency under noise, Gaussian blur, and contrast perturbation.

C. Segmentation Results

The Attention U-Net segmentation model achieves a Dice score of 0.893 and IoU of 0.902 on the test set, with the estimated tumor area for the reference scan at 308 mm². Table III summarizes segmentation metrics across all four U-Net variants.

Architecture	Dice Score	IoU	Precision	Recall
U-Net (Baseline)	0.861	0.875	0.843	0.880
Residual U-Net	0.878	0.889	0.869	0.891
Multi-modal U-Net	0.882	0.891	0.874	0.895
Attention U-Net ★	0.893	0.902	0.887	0.901

Table III. Segmentation metric comparison across U-Net variants. ★ indicates recommended default model.

D. Model Comparison Summary

Across all experiments, the Vision Transformer demonstrates a clear accuracy advantage, consistent with its capacity to model global spatial relationships via self-attention. However, its 86 million parameter count and 2.5-second inference time may constrain real-time deployment in resource-limited settings. The Transfer Learning model (ResNet-50 fine-tuned) represents the optimal balance of performance and computational cost and is designated the Recommended model in the dashboard. The CNN is best suited for high-throughput pre-screening applications where classification speed is the primary constraint.

These findings suggest a tiered deployment strategy: CNN for first-pass screening, Transfer Learning as the primary diagnostic aid, and Vision Transformer for second-opinion review in cases of high clinical uncertainty.

VI. DISCUSSION

The NeuroLens AI results contribute several observations relevant to the applied medical imaging community. First, the systematic per-scan multi-model comparison exposed meaningful confidence divergence across architectures on individual MRI scans: in approximately 8% of test cases, the CNN and ViT yielded different binary predictions with overlapping confidence ranges. This class of uncertainty is not visible in aggregate benchmark tables and motivates the inclusion of per-scan ensemble disagreement as an explicit uncertainty indicator.

Second, the Grad-CAM visualizations corroborate the classification decisions for the large majority of correctly classified scans, adding qualitative support for the predictions. However, for misclassified scans, the heatmaps frequently show diffuse, non-focal activation patterns, suggesting that classification errors are associated with spatially incoherent feature usage—a finding consistent with prior observations in medical imaging Grad-CAM studies [15]. This provides a practical screening heuristic: diffuse heatmaps should trigger elevated clinical scrutiny.

Third, the Attention U-Net's superiority over the baseline U-Net (Dice 0.893 vs. 0.861) confirms that attention gating meaningfully improves boundary precision for small and irregular tumor masses, consistent with Oktay et al. [19]. The IoU advantage (0.902 vs. 0.875) is particularly notable, as IoU penalizes false positives and false negatives symmetrically and is more stringent than Dice for small-area tumors.

A key limitation of this work is that reported metrics are derived from a single publicly available dataset. Generalization to out-of-distribution scanner protocols, contrast enhancement variations, or rare tumor subtypes has not been evaluated. Furthermore, the segmentation ground-truth masks used for training were generated from semi-automated tools and may contain annotation noise. Clinical deployment would require validation against expert radiologist annotations across diverse acquisition settings.

VII. LIMITATIONS AND FUTURE WORK

Several limitations of the current framework warrant future investigation. The binary (tumor / normal) classification formulation does not distinguish between tumor histological subtypes such as glioma, meningioma, pituitary adenoma, or metastatic lesions, each of which carries different clinical implications and treatment pathways. Extension to multi-class classification is a direct next step. Additionally, the present system operates on 2D axial slices; volumetric 3D MRI analysis would capture inter-slice tumor extent and morphology unavailable from single-plane views.

From an architectural standpoint, hybrid CNN-Transformer models that combine the local feature extraction strengths of convolutional layers with the global context modelling of self-attention represent a promising direction for improving both accuracy and inference efficiency. Federated learning protocols would enable multi-institutional model training without sharing raw patient data, directly addressing data privacy constraints that currently limit large-scale medical AI research.

The explainability module could be extended beyond Grad-CAM to include Guided Grad-CAM, Integrated Gradients, or SHAP-based attribution methods, providing richer and more faithful explanations. Integration with hospital PACS infrastructure and support for DICOM-RT (radiotherapy planning) exports would improve clinical workflow compatibility.

VIII. CONCLUSION

This paper presented NeuroLens AI, a comprehensive deep learning framework for brain tumor detection, model comparison, explainability, and segmentation from MRI scans. By integrating CNN, Transfer Learning, and Vision Transformer architectures within a unified per-scan comparative pipeline, the system moves beyond single-model accuracy benchmarks to expose confidence, uncertainty, and robustness information that is practically meaningful for clinical decision support.

The Vision Transformer achieved the highest classification accuracy (95.1%) and ROC-AUC (0.987), while the CNN provided the fastest inference (0.5 s) at 88.4% accuracy—a trade-off quantified for the first time in a unified comparative interface. Grad-CAM heatmaps provided spatial explanations for model decisions, and the Attention U-Net delivered strong tumor boundary segmentation (Dice 0.893, IoU 0.902). The complete framework, including training scripts, evaluation utilities, and interactive dashboards, is publicly available and designed for reproducibility and extensibility.

NeuroLens AI is not a certified clinical diagnostic system and all outputs must be reviewed by qualified medical professionals. It is intended as a research platform for developing, comparing, and explaining brain tumor detection models, and as a demonstration of how multi-model explainable AI can be delivered through a practical interface accessible to both researchers and clinicians.

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